

3. A student examined cells from an onion epidermis using a light microscope.

The student used the following method:

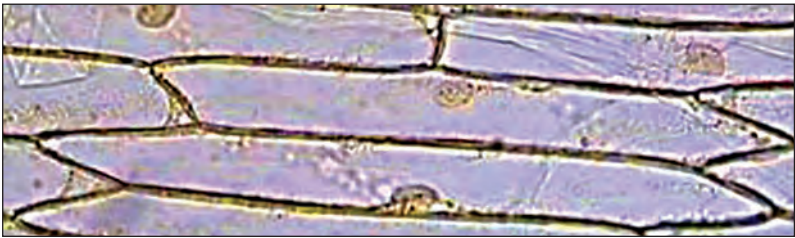
- 1. Peel a thin layer of epidermis from inside a freshly cut onion.
- 2. Lay epidermis on glass slide.
- 3. Add a drop of iodine solution to the onion epidermis on the glass slide.
- 4. Gently lower a coverslip onto the glass slide.
- 5. Using a light microscope examine the slide to identify some of the cell structures.

(a) One hazard for this method has been assessed and given in the table below. Identify **one** further hazard involved in this method and complete the risk assessment for it in the table below. [2]

Hazard	Risk	Control measure
Iodine solution is an irritant	Transfer of iodine solution from hand to eye can irritate the eyes	Use lowest concentration possible
.....		
.....		



(b) A photograph of the student’s observation is shown below.



(i) Draw **one** of the onion cells from the photograph above in the space below. [1]

(ii) **On your drawing**, label **three** parts of the cell **you can see** in the photograph. [2]

(c) State how the student would calculate the total magnification of the microscope she was using. [1]

.....

.....

6

